

TECHNICAL REPORT

Two gaps in the cv32e40p-dv-review Verilator core testbench, with patches

openhwgroup/cv32e40p

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1 Answer

The Verilator core testbench in *cv32e40p-dv-review* (commit 89ad543) has two gaps. Two small patches close them. Each applies cleanly on its own, and they apply together with the FPU patch first.

Gap	Effect	Patch
The FPU latency and Zfinx parameters are hard-coded to 0.	The pipelined configurations (F1, F2, Z1, Z2) and the Zfinx configurations can't be selected. Every build gets FPU latency 0 and ZFINX 0, whatever the user asks for.	patches/tb_fpu_config_plumbing.patch
The core's debug_req_i is tied to 0.	A debug request from the testbench's virtual peripheral never reaches the core. A directed debug test ran to completion while none of its 32 debug requests reached the core.	patches/tb_debug_req_wiring.patch

2 The question

Our work on #1060 and the FPU pipeline depth needed the testbench to build F1, F2, Z1 and Z2. Our work on #1004 needed debug requests to reach the core. Both failed silently: the testbench built and ran, but not with the configuration or stimulus we intended. This report records the two causes so the test suite can be trusted on these configurations.

3 What we did

1. We read the testbench source at 89ad543 and found where the parameters and the debug request are tied off.
2. We wrote one patch per gap and applied each to a clean checkout: each on its own, and both together (FPU patch first).
3. We checked that the parameters and the debug request now reach the core, using run-time effects (below) rather than reading the code.
4. We reran the ACT4 anchors with the patches applied to confirm nothing else changed.

4 Results

Gap 1: FPU latency and Zfinx parameters

At 89ad543:

File:line	Code	Problem
tb/core/cv32e40p_dut_wrap.sv:97-99	.FPU_ADDMUL_LAT (0), .FPU_OTHERS_LAT (0), .ZFINX (0)	Constants in the v1.8.3 instance of the core wrapper
tb/core/cv32e40p_dut_wrap.sv:48	.PULP_ZFINX (0)	The same, in the older v1.0.0 instance

File:line	Code	Problem
tb/core/tb_top.sv:25	parameter FPU_EN = 0	tb_top exposes only FPU_EN, so there is no way to pass the other parameters down

The patch adds FPU_ADDMUL_LAT, FPU_OTHERS_LAT and ZFINX as parameters of tb_top and cv32e40p_dut_wrap, and passes them through to the core. The defaults stay 0, so existing builds don't change.

Evidence that the parameters now reach the FPU:

Check	Result with the patch
ACT4 end-of-test simulation time against F0	Differs on F1 in 7 of 180 passing tests and on F2 in 180 of 180. On F1 the extra add/multiply stage is mostly absorbed by the write-back stage, which is why most F1 times equal F0.
Cycle counts of FP kernels (report 01)	Change with latency exactly as the pipeline predicts, for example a dependent dot product: F0 4371, F1 4627, F2 5141 cycles
ACT4 pass counts	Unchanged by the patch: I0 94/94, F0 180/181 (only priv/SmF/SmF-00 fails, which is #1060)

Gap 2: debug request tied to 0

At 89ad543:

File:line	Code	Problem
tb/core/cv32e40p_dut_wrap.sv:85 and :127	.debug_req_i (1'b0)	The core's debug request is a constant 0 in both core instances
tb/core/tb_top.sv:276	.debug_req_o ()	The virtual peripheral's debug request output (mm_ram) is left unconnected

The patch adds a debug_req_i port to cv32e40p_dut_wrap, a debug_req net in tb_top, and connects mm_ram.debug_req_o to it.

Evidence, from counters bound into the design (I0, directed test first_fetch_dt for #1004, which requests 32 debug entries):

Counter	With the patch
Debug requests written by the program to the testbench	32
Debug request rising edges at the core's debug_req_i	32 (64 cycles high)
Debug entries taken by the core	32
Hits on the #1004 condition	1

Without the patch, the same counters showed the program running to the end while none of its 32 debug requests reached the core: 0 cycles of debug_req_i high, and 0 hits on #1004. Those numbers come from our log of that diagnosis; the run's record was later overwritten, so only the patched run's record is kept. With the patch, ACT4 on I0 is still 94/94 and F0 is still 180/181 (only SmF-00 fails).

5 What this means for TRL5

- The pipelined configurations can't be verified with this testbench as it stands. Any F1, F2, Z1 or Z2 result produced with it is really an F0 or Z0 result. This matters for the FPU-depth decision ([report 01](#)) and for #1060 ([report 02](#)).
- Debug-request tests in this testbench never deliver a request to the core. They can pass without exercising the core's external debug-request path (debug entry by ebreak is not affected). Any TRL5 claim that rests on debug tests run in this testbench should be rechecked with the wiring patch applied.
- Both patches are small and keep existing behaviour by default. We are happy to open them as pull requests against cv32e40p-dv-review if that is useful.

6 How to repeat

```
git clone https://github.com/openhwgroup/cv32e40p-dv-review
cd cv32e40p-dv-review
git checkout 89ad543
git apply /path/to/core-v-investigation-reports/cv32e40p-trl5/patches/tb_fpu_config_plumbing
```

```
.patch
git apply /path/to/core-v-investigation-reports/cv32e40p-trl5/patches/tb_debug_req_wiring.patch
```

Then build the Verilator core testbench with the parameters overridden on the Verilator command line, for example -GFPU_EN=1 -GFPU_ADDMUL_LAT=1 -GFPU_OTHERS_LAT=1 for F1, or -GFPU_EN=1 -GZFINX=1 for Z0.

To see gap 1 for yourself, run any FP-heavy program on F0 and on F2. The end-of-test time:

- is identical without the patch;
- differs with the patch.

To see gap 2 for yourself, run a program that writes the testbench's debug-control register, then count debug-mode entries in the trace or with dcsr. You get none without the patch, and one per request with it.

Tools: Verilator 5.050.

7 Limits

- Gap 2 was observed with one directed program. We didn't audit every debug test in the suite for whether it was relying on the tie-off.
- There is no recorded run of the unpatched build of gap 1 on F1 or F2. The evidence that the patch is needed is the source (the constants above). The evidence that it works is the changed timing and cycle counts.
- The irq_i wiring is not part of these gaps. It is connected in this testbench. A separate problem with irq_i tied to 0 exists in the older core-v-verif core testbench and is not covered here.

8 Files

- This report: cv32e40p-trl5/04-testbench-gaps.md
- patches/tb_fpu_config_plumbing.patch: FPU latency and Zfinx parameters, tb_top.sv and cv32e40p_dut_wrap.sv
- patches/tb_debug_req_wiring.patch: debug request from mm_ram to the core, tb_top.sv and cv32e40p_dut_wrap.sv